Design and Development of an Interactive Metaverse-Based Virtual Cricket Stadium with Walk-Through and Meta Playing Modes Using Blender

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Course: Metaverse

***Abstract*—This course portfolio presents the structured design and development of an interactive metaverse-based virtual cricket stadium developed using Blender. The project focuses on building a realistic digital twin of a cricket stadium integrated with two operational modes: Walk-Through Mode for immersive exploration and Meta Playing Mode for interactive engagement. The development process was carried out over fourteen weeks, including research, 3D modeling, character integration, and environment optimization. The project demonstrates metaverse concepts such as immersive environments, spatial presence, digital twin systems, and interactive virtual infrastructure.**

I. Introduction

The Metaverse is an immersive virtual ecosystem that enables users to interact within 3D digital environments. A core concept within the metaverse is the creation of digital twins, which are virtual replicas of real-world infrastructure.

This project aims to develop a realistic cricket stadium as a digital twin using Blender. The system integrates immersive exploration and interactive elements, providing a structured representation of metaverse architecture principles.

II. Literature Review

The development of immersive virtual environments has been widely explored in recent research. Nendya et al. [1] discussed the metaverse as a game ecosystem, focusing on player immersion and interaction analysis within 3D environments. Their work emphasizes the importance of spatial experience and engagement, which directly supports our walk-through design approach.

Mystakidis [2] highlights immersive learning and digital twin environments as key pillars of metaverse systems. Lee et al. [3] provide a comprehensive overview of metaverse evolution and architectural frameworks. Unity Technologies [4] outlines interactive environment development techniques relevant to immersive system design.

III. Project Objectives

- Design a realistic 3D cricket stadium.
- Implement immersive walk-through navigation.
- Develop interactive meta playing mode.
- Integrate characters within the virtual stadium.
- Demonstrate digital twin and metaverse principles.

IV. Development Progress (Weeks 1–7)

A. Week 1: Research and Planning

Metaverse concepts, digital twin systems, and immersive design research were studied. Project scope and system architecture were finalized.

B. Week 2: Stadium Outer Structure

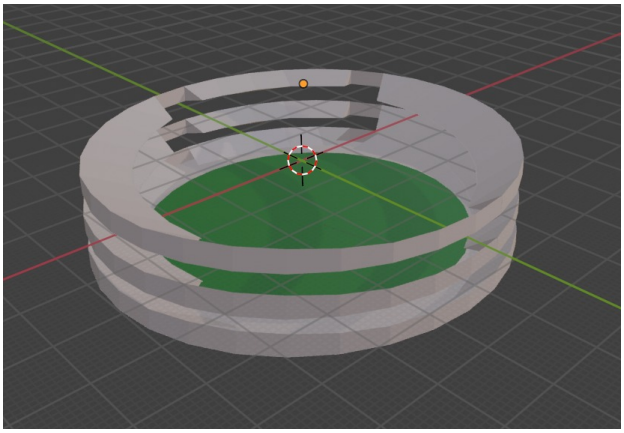

Fig. 1. Stadium Outer Structure Modeling

The circular stadium framework was modeled using mesh tools, scaling adjustments, and structural alignment techniques.

C. Week 3: Stand Development

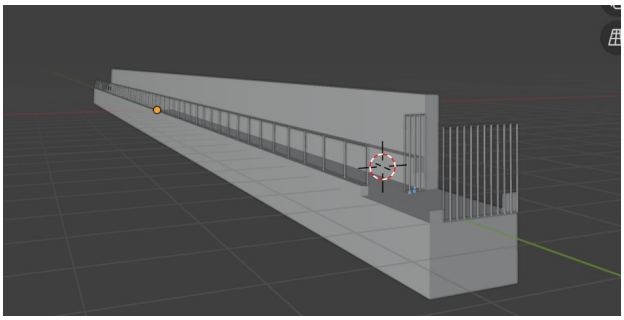

Fig. 2. Tier-Based Stand Modeling

Tier-based stands were created using duplication and array modifiers to ensure uniform seating structure.

D. Week 4: Pitch Modeling

The pitch and boundary layout were modeled with accurate proportions and grass textures applied.

E. Week 5: VIP Box Design

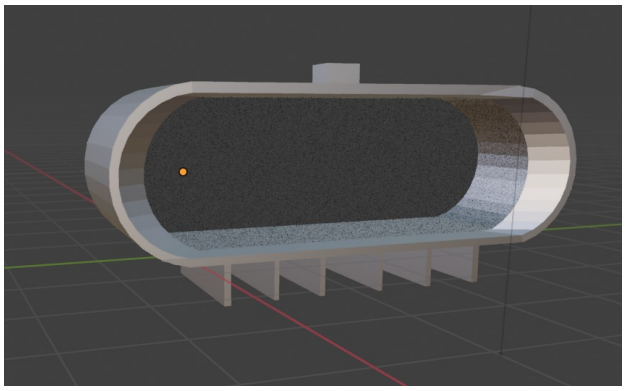

Fig. 3. VIP Box Structure

A premium enclosed VIP section was designed with transparent materials and seating layout.

F. Week 6: Seating and Chair Modeling

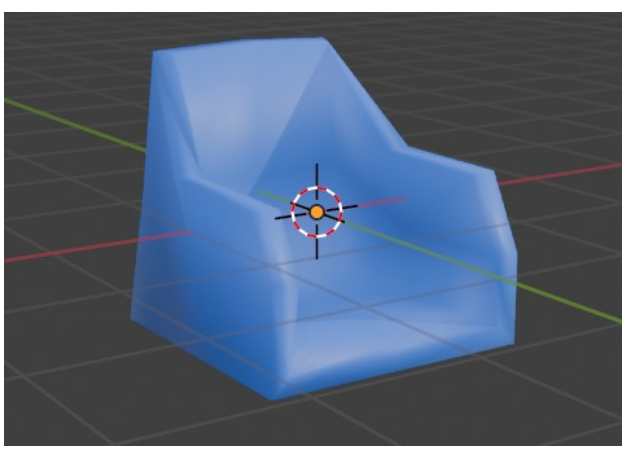

Fig. 4. VIP Chair Design

Individual chair models were created and arranged systematically using duplication tools.

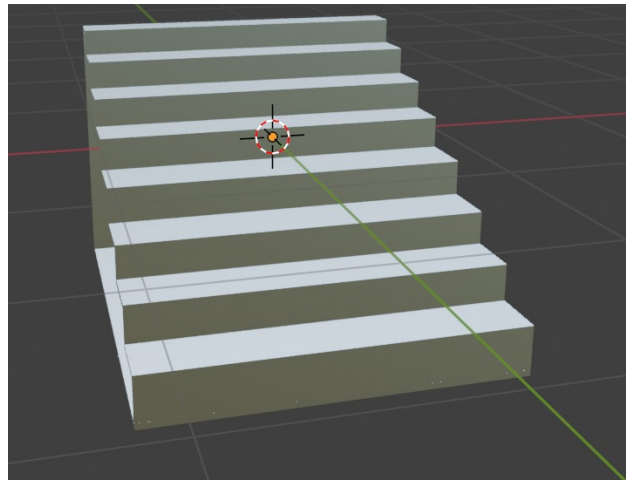

Fig. 5. Audience Sitting Area Arrangement

Audience seating areas were aligned for structural consistency and realism.

G. Week 7: Character Integration

Basic humanoid characters were placed within the stadium environment to simulate audience presence and prepare for interactive functionality.

V. System Architecture

The system consists of two operational modules:

A. Walk-Through Mode

- First-person camera navigation.
- Free exploration of stadium sections.
- Real-time rendering environment.

B. Meta Playing Mode

- Interactive character placement.
- Simulated gameplay environment.
- Scoreboard and virtual interaction integration (planned).

VI. Future Development Plan (Weeks 8–14)

- Implement full first-person navigation controls.
- Integrate interactive scoreboard system.
- Add stadium floodlight system.
- Introduce day/night rendering modes.
- Optimize polygon count and rendering performance.
- Explore VR compatibility.
- Final testing and documentation.

VII. Challenges Faced

- Managing large-scale 3D environments.
- Maintaining realistic proportions.
- Lighting optimization.
- Rendering performance constraints.

VIII. Learning Outcomes

- Advanced 3D modeling skills.
- Understanding of digital twin architecture.
- Immersive environment design principles.
- Metaverse system integration knowledge.

IX. Conclusion

The project demonstrates the structured development of a metaverse-based virtual cricket stadium incorporating immersive walk-through and interactive meta playing modes. The integration of digital twin concepts, environment design, and structured weekly development reflects a comprehensive understanding of immersive virtual infrastructure systems.

References

[1] M. B. Nendya, I. K. D. Senapartha, H. B. Santoso, R. Delima, M. Wibisono, J. A. Antono, and L. Wuttisittikulkij, "The Metaverse as a Game Ecosystem: Evaluating Player Interaction Through Mini-Game Integration and Experience Analysis," Universitas Kristen Duta Wacana, Yogyakarta, Indonesia.

[2] S. Mystakidis, "Metaverse," Encyclopedia, vol. 2, no. 1, pp. 486–497, 2022.

[3] L.-H. Lee et al., "All One Needs to Know about Metaverse: A Complete Survey on Technological Singularity, Virtual Ecosystem, and Research Agenda," Journal of Latex Class Files, 2021.

[4] A. Juliani et al., "Unity: A General Platform for Intelligent Agents," arXiv preprint arXiv:1809.02627, 2018.